Multifrequency Analysis: 50×50 Cell Model
(notes)

1 Overfitting in Single-Frequency Optimization

Neural network optimization of the cloak material parameters can easily achieve excellent cloaking performance at a single design frequency. However, optimizing at one frequency produces a highly frequency-specific solution that fails to generalize to other frequencies.

Figure 1 illustrates this phenomenon by comparing two optimization runs on identical 50×50 cell models, trained at different frequencies (2.5 kHz and 3.0 kHz). Each model is then evaluated across the range 1.0–4.0 kHz.

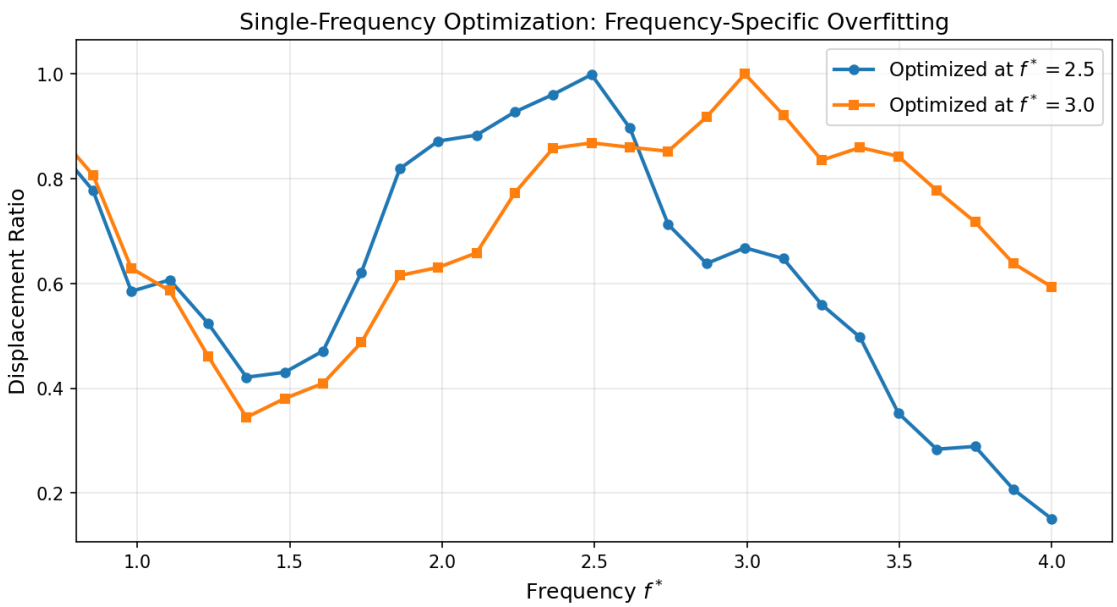

Figure 1: Frequency response of two independently optimized 50×50 cell models, each trained at a single frequency (2.5 kHz and 3.0 kHz). The sharp minima at the training frequencies illustrate frequency-specific overfitting; performance away from the design frequency degrades significantly.

2 Multifrequency Objective

To address narrow-band overfitting, we adopt a multifrequency training objective that enforces broadband cloaking performance across a range of frequencies. The loss function is modified to evaluate cloaking distortion at multiple frequencies and sum their contributions, ensuring that the optimization does not trade performance at one frequency for another. Figure 2 shows the cloaking performance of a single 50×50 cell model optimized using the multifrequency objective.

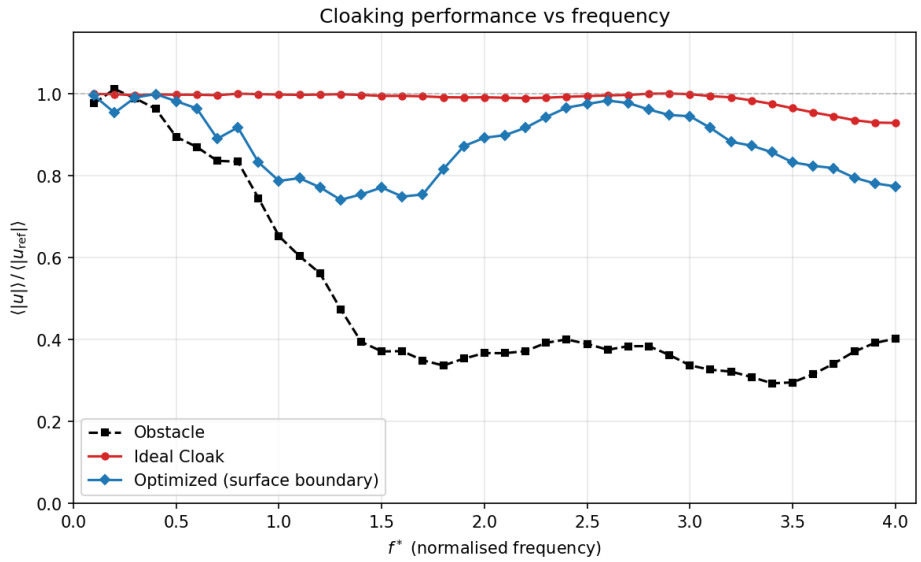

Figure 2: Frequency sweep of the multifrequency-optimized 50×50 cell model.

3 Floquet–Bloch Analysis

The multifrequency-optimized material field is analyzed through Floquet–Bloch band structure computation at the optimal parameters (Figure 3).

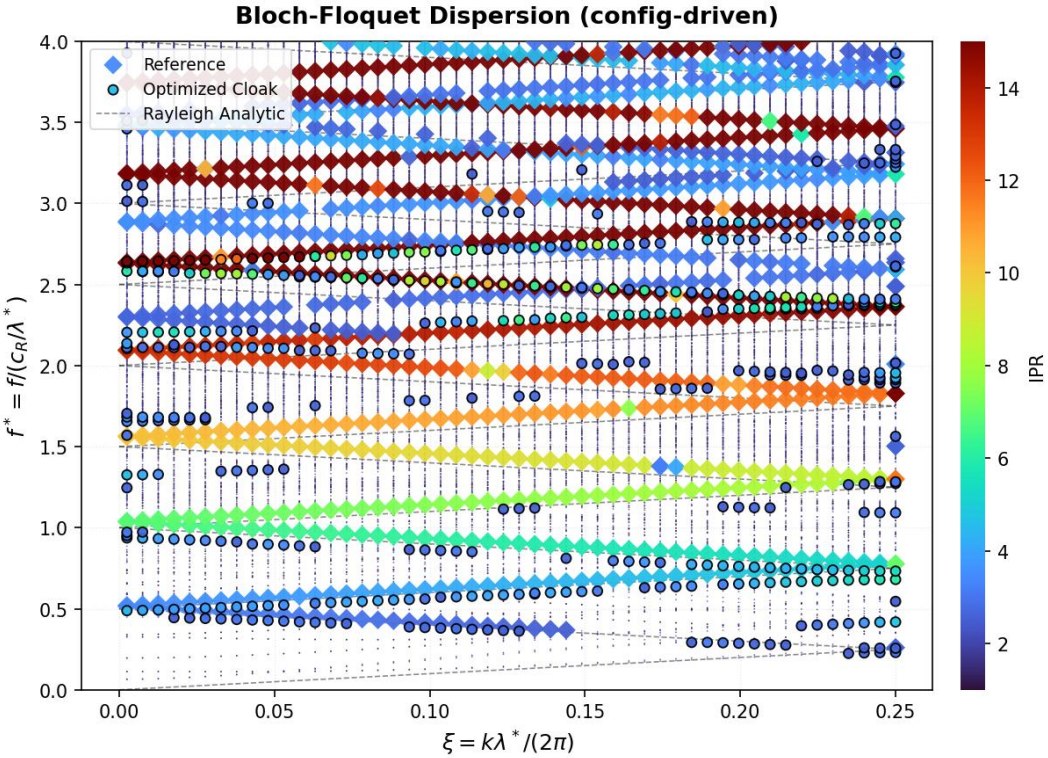

Figure 3: Floquet–Bloch band structure analysis of the multifrequency-optimized 50×50 cell model. Best cloaking for these parameters corresponds to $f \approx 2.6$. IPR cutoff $= 2.5$.